STUDENT REFERENCE & INTERNAL USE NOTICE

This protocol documents standard in-vitro autoaggregation and coaggregation assay procedures for probiotic lactic acid bacteria (LAB) used in HOPE Lab research and is PROVIDED ONLY FOR THE HOPE LAB MEMBERS for training and internal reference. External users should independently assess method suitability, verify current standards and chemical safety for reagents, and obtain required biosafety authorization and other essential approvals. This document does not replace an institutionally approved SOP or method-specific risk assessment.

TITLE

Autoaggregation and Coaggregation Assays for Probiotic Lactic Acid Bacteria

PRINCIPLE

Aggregation is a phenotypic property in which microbial cells associate with one another and settle out of suspension. Autoaggregation describes the self-association of cells belonging to the same bacterial strain, whereas coaggregation describes the physical association between cells from two distinct bacterial strains (e.g., a probiotic candidate and a target pathogen). These properties are evaluated spectrophotometrically by monitoring optical density (OD) in a defined upper portion of an undisturbed cell suspension over time. A progressive reduction in measured upper-layer absorbance reflects cellular settling; however, readings are also influenced by cell morphology, chain or clump formation, buffer ionic strength, and measurement geometry.

These assays serve as standard in-vitro screening tools for candidate probiotic lactic acid bacteria (LAB). They provide descriptive phenotypic data and do not, by themselves, establish epithelial cell adhesion, mucosal colonisation, intestinal persistence, or probiotic efficacy in vivo.

Learning Outcomes

- Explain the mechanistic distinction between strain-specific autoaggregation and inter-strain coaggregation.
- Standardize initial bacterial suspensions to uniform starting optical densities and include time-matched controls.
- Calculate percentage autoaggregation and coaggregation accurately from blank-corrected absorbance measurements.
- Distinguish independent biological replicates from technical repeat measurements.
- Interpret the physical and biological limitations of optical-density-based aggregation assays.

OBJECTIVE & SCOPE

Objective: To describe the standardized principles, mathematical calculations, experimental controls, and reporting requirements for evaluating in-vitro autoaggregation and coaggregation phenotypes in probiotic lactic acid bacteria (LAB).

Scope: Applicable to planktonic lactic acid bacteria (e.g., *Lactiplantibacillus, Limosilactobacillus, Pediococcus, Streptococcus, Weissella* spp.) and co-tested indicator strains for which a validated laboratory SOP is available. Strains that naturally form dense chains, clumps, or non-uniform suspensions require validated dispersion procedures or

explicit methodological limitation statements. This assay does not measure host epithelial adhesion or in-vivo probiotic colonisation.

CORE CONCEPTS & CALCULATIONS

- **Optical Density as an Operational Proxy:** OD measures light scattering in the optical path. A decline in upper-layer OD indicates cell sedimentation, but light scattering is also affected by cell size, chain morphology, and cuvette path length. OD is therefore an operational proxy rather than a direct count of cellular aggregates.
- **Autoaggregation Formula:** Let A_o represent the blank-corrected initial absorbance of a standardised single-strain suspension at time t = 0, and A_t represent the blank-corrected absorbance of the undisturbed upper layer at time t (e.g., 2 h, 4 h, 24 h):
  Autoaggregation (%) = $[(A_o - A_t) / A_o] \times 100$
- **Coaggregation Formula:** For strains x and y, let $A_x(t)$ and $A_y(t)$ be the blank-corrected absorbances of the separate monoculture suspensions at time t, and $A_{xy}(t)$ be the blank-corrected absorbance of their mixed suspension (1:1 ratio) at the same matched time point:
  Coaggregation (%) = $\{[(A_x(t) + A_y(t)) / 2 - A_{xy}(t)] / [(A_x(t) + A_y(t)) / 2]\} \times 100$
  Note: If the denominator approaches zero or falls below the instrument's linear detection limit, the value is uninterpretable and must not be reported. Negative coaggregation percentages can arise when the mixed suspension retains higher turbidity than the mean of the time-matched monocultures; verify raw readings and blank corrections before retaining valid negative values.
- **Mandatory Experimental Controls:** A sterile buffer/matrix blank establishes baseline background absorbance. Separate monoculture suspensions of both partner strains (x and y) must be assayed concurrently at identical time points to account for independent autoaggregation.

MATERIALS & EQUIPMENT

Reagents & Consumables

- Selective/enrichment media (e.g., MRS broth/agar for LAB; BHI/TSB for indicator strains)
- Defined sterile suspension buffer (e.g., Phosphate-Buffered Saline, PBS, pH 7.2–7.4)
- Matched optical cuvettes (1 cm path length) or sterile 96-well microplates
- Sterile centrifuge tubes (15 mL / 50 mL) and micropipette tips
- Reference LAB strains and indicator organisms (with documented provenance)

Equipment & Instrumentation

- Calibrated UV-Vis Spectrophotometer or Microplate Reader (OD_{600} nm)
- Refrigerated Centrifuge (capable of 4,000–6,000 × g at 4°C)
- Bacteriological Incubator (calibrated to 37 ± 1°C)
- Vortex Mixer for uniform initial suspension homogenization
- Biosafety Cabinet (Class II BSL-2) & Precision Timer

STEP-BY-STEP PROCEDURE

Step 1: Culture Preparation & Cell Harvesting

Inoculate purified LAB isolates and coaggregation partner strains into their respective validated broth media (e.g., MRS broth for LAB) and incubate at 37°C for 18–24 hours to reach late-exponential/early-stationary phase. Harvest cells by centrifugation (e.g., 5,000 × g for 10 min at 4°C). Discard the supernatant and wash the cell pellet twice with sterile PBS buffer (pH 7.2–7.4) to remove residual medium components.

Step 2: Standardisation of Starting Bacterial Suspensions

Resuspend washed cell pellets in sterile PBS buffer. Vortex thoroughly for 10–15 seconds to obtain a uniform suspension. Adjust the cell density of each individual strain spectrophotometrically at 600 nm (OD_{600}) to a standardized baseline (typically OD_{600} = 0.50 ± 0.05 or 1.00 ± 0.05, as defined by the study SOP) using sterile PBS as the blank. Record the exact blank-corrected initial absorbance (A_o) for each preparation.

Step 3: Autoaggregation Assay Setup & Incubation

Transfer an aliquot (e.g., 4.0 mL) of each standardized single-strain suspension into sterile, matched glass or polystyrene tubes. Vortex gently for 10 seconds to ensure initial homogeneity, record the t = 0 absorbance (A_o), and incubate the tubes upright and completely undisturbed at the specified assay temperature (e.g., 37°C or room temperature) for defined time intervals (e.g., 1, 2, 4, and 24 hours).

Step 4: Coaggregation Assay Setup & Incubation

Combine equal volumes (e.g., 2.0 mL each, 1:1 v/v ratio) of the standardized probiotic LAB suspension (strain x) and partner strain suspension (strain y) into a sterile tube. Vortex for 10 seconds to mix thoroughly. In parallel, prepare separate monoculture control tubes containing 4.0 mL of strain x alone and 4.0 mL of strain y alone. Incubate all mixed and monoculture tubes upright and undisturbed under identical temperature and time conditions.

Step 5: Upper-Layer Sampling & Optical Density Measurement

At each predetermined time point (t), carefully aspirate a defined small volume (e.g., 0.1–0.5 mL) from the upper surface layer of each undisturbed suspension without agitating or resuspending the settled pellet. Transfer to a cuvette (diluting in a consistent, documented ratio with PBS if required to stay within linear range) and measure OD_{600} against the sterile PBS blank. Record raw absorbance values for monocultures $A_x(t)$, $A_y(t)$, and mixed suspensions $A_{xy}(t)$.

Step 6: Data Processing, Blank Correction & Replication

Subtract the matched buffer blank from all raw absorbance readings. Calculate Autoaggregation (%) and Coaggregation (%) using the defined formulas. Perform at least three independent biological replicates (from separate starter cultures) with technical duplicate/triplicate readings per assay, and report mean values alongside standard deviation (SD) or standard error.

CRITICAL FACTORS AFFECTING RESULTS

Variable	Effect of Inadequate Control	Required Control Standard
Culture Growth State	Cell-surface proteins and EPS vary across growth phases	Standardise incubation time (18–24 h) and growth stage
Starting OD_{600}	Inconsistent initial density distorts percentage settling rates	Standardise every preparation to exact target OD; log A_o
Cell Morphology	Natural chains or clumps alter light scattering independently	Document cell morphology microscopically; note limitations
Buffer Composition	Variations in pH or ionic strength alter surface charge interactions	Use defined sterile PBS (pH 7.2–7.4); record lot/pH
Incubation Temperature	Brownian motion and cell-surface interactions vary with temp	Maintain pre-specified, monitored temperature (e.g., 37°C)
Measurement Geometry	Sampling from different depths changes measured turbidity	Aspirate fixed volume from exact upper meniscus layer

Physical Disturbance	Vibrations or jarring resuspend settled cell aggregates	Keep tubes completely undisturbed; handle gently at readouts

QUALITY CONTROL & TROUBLESHOOTING

Quality Control (QC): Include sterile buffer blanks and time-matched monoculture controls for every coaggregation pair. Include characterized high- and low-aggregating reference LAB strains when available. Retain raw absorbance values, blank corrections, dilution factors, and biological vs. technical replicate metadata - never report final percentages alone.

Observation	Possible Explanation	Corrective Action
Little or no autoaggregation	Strain exhibits low settling phenotype or buffer is unsuitable	Verify raw OD data & reference control; report low phenotype objectively
Negative coaggregation (%)	Mixed suspension retains higher OD than mean monocultures	Audit time-matched controls, blank subtraction & math; retain valid negatives
High replicate variability	Inconsistent starting OD, sampling depth, or tube disturbance	Standardise initial OD_{600} strictly; avoid agitating tubes during sampling
Unstable blank readings	Buffer contamination, scratched cuvettes, or lamp drift	Use fresh sterile PBS; inspect cuvettes; re-zero spectrophotometer
Non-uniform suspension	Strain naturally forms macroscopic clumps or long chains	Vortex consistently prior to t = 0; note morphological limitation in report

BIOSAFETY & ADDITIONAL CONSIDERATIONS

- **Biosafety & Containment:** Handle all bacterial cultures, especially when co-testing probiotic LAB against opportunistic or enteric pathogen, under institutional BSL-2 containment inside a certified Biosafety Cabinet. Autoclave all contaminated tubes, cuvettes, and pipette tips at 121°C for 30 minutes prior to disposal.
- **Scientific Interpretation Boundary:** Report autoaggregation and coaggregation strictly as in-vitro physicochemical surface phenotypes under the stated buffer, temperature, and time conditions. Do not equate high in-vitro aggregation with intestinal epithelial adhesion, mucosal colonisation, pathogen exclusion in vivo, or clinical probiotic efficacy.

REFERENCES & RELATED DOCUMENTS

Paul S, Hossain TJ, Ali F, Hossain ME, Chowdhury T, Faisal IK, Ferdouse J. (2024). Assessment of the in-vitro probiotic efficacy and safety of *Pediococcus pentosaceus* L1 and *Streptococcus thermophilus* L3 isolated from Laban, a popular fermented milk product. Archives of Microbiology, 206, 82. doi:10.1007/s00203-023-03812-5

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